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EMERGING SYSTEMIC TURN IN SERVICE DESIGN

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1 Introduction

A substantial effort of research and educational programmes in service design has focused on providing a methodological approach and methodologies that could qualify students to work in interdisciplinary teams. This has included experts in similar disciplines, such as industrial design, interaction design, management or process engineering (Holmlid, 2007; Morelli, 2009). Around the 1990s, service design started to be viewed as a separate research domain in design academia when the Köln International School of Design introduced service design courses (e.g. Hollins & Hollins, 1991; Sun, 2020). Service design education shaped an *operative paradigm* (Arbnor & Bjerke, 1997) that helped existing professionals include then-new user-centred, design-oriented, participatory methods and methodologies in their practices. This chapter proposes that service design research and education have grown into a new era, and that the questions they face are even more complex than they were in the last turn.

Service design in the educational context has been supported by research (e.g. Huan, Arvola, & Holmlid, 2020; Meroni & Sangiorgi, 2016; Sangiorgi & Pacenti, 2010). This has shifted the attention from a very hands-on focus typical for service design practice, especially in the early years of service design. There is already an extensive body of service design research initiated by the same scholars who teach in major service design programmes. Service design education is, by far, not only informed by practice but also by a strong research orientation. Service designers are educators and scholars who have initiated the first academic books on service design (Polaine et al., 2013; Stickdorn & Schneider, 2012).

The maturation of academic research rigour and the need for defining a professional profile for service designers often led to the development of technical

DOI: 10.4324/9781003501039-5
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and methodological approaches to service design. However, the idea that design, as a discipline, could assume a *technical* positioning that only considers the professional role (i.e. functional, technical, or organisational aspects of its action) is proving to be wrong or at least insufficient to address the significant sociotechnical changes generated by the major crises in our society. Design research has already highlighted the need for design to take a position concerning the impact of major production and consumption systems (Norman & Euchner, 2023) and problems, such as migration, inclusion and gender issues, and the right of different design cultures to express their potential, outside a colonising umbrella of the Global North (Light, 2019). This chapter asks if a new turn in service design research and education is emerging.

The question about a paradigm shift was discussed by Thomas Kuhn (1962) in his book *The Structure of Scientific Revolutions*. Kuhn suggests that the dominant active paradigm is usually characterised by theories and tools for approaching certain problems. This can be challenged by anomalies that push the boundaries of what is known, and new theories and exploration are needed to explain the anomalies. During the shift, the new paradigm would encounter resistance before becoming normal.

While design research and experimentation look at this perspective change, service design education has often held a conservative position (Becermen & Simeone, 2019). It has focused on a profile of designers as problem solvers. In contrast, the hypothesis of a different engagement in society requires future designers to understand and learn how to work with conflicts, tensions, agonism, chaos and complexity (Ehn, Nilsson et al., 2014).

The cultural evolution for social change and increasing sensitivity to the emerging crisis find fertile ground in the increasing interest of several universities in cultivating a *third mission* (the first two missions being education and research) that focuses on the civic role of universities and their contribution to society (Compagnucci & Spigarelli, 2020). This perspective encourages universities to project their educational and research activities over their physical and social surroundings. In some European countries, especially in the Nordics, this third mission is discussed in the context of industry-university collaboration, and the industry relevance in teaching has already been discussed for some decades (Kivinen & Nurmi, 2014; Ranga, Perälampi, & Kansikas, 2016). This resonates well with the role of service design in adding value by applying, for example, more human-centred methods (Sangiorgi et al., 2019; Yu & Sangiorgi, 2018).

This perspective implies that service design education looks beyond the existing professional horizon and considers the role service design can have, for instance, in social innovation, in the public sector, or in the definition of policies that address emerging social (Staszowski, Brown, & Winter, 2016), urban (Villari, 2022) or health-related issues (Jones, 2013). Kuzmina and Bhamra (2014) studied service design as a transformative approach in education for sustainable development. Miettinen et al. (2022) studied how critical thinking and sustainability

can be included in a service design teaching model. Their design teaching model proposes more thematic study areas, such as strategic service design, digital service design and place-specific service design. Furthermore, some researchers have initiated research on the strategic role of service design in organisations (Foglieni et al., 2017). However, these efforts are a minor part of the vast research endeavour to discover the benefits of adopting service design methods and processes. Some scholars discuss service design in the context of more complex societal challenges, service systems and sustainability (Jones et al., 2014; Manzini & Vezzoli, 2003; Santos & Sustar, 2023; Van der Bijl-Brouwer, 2017).

Service design is evolving with the social, economic and technical systems with which it interacts, and this context is now severely challenged by multiple crises concerning environmental aspects, social inequalities, wars, migrations and deep cultural changes. This chapter discusses the turn in the service design paradigm from the micro-level operational and methodological practice towards designing at the macro level, which concerns more complex and strategic systems. This turn is discussed first through a literature review, then with service design researchers in a co-design workshop and verified through data collected in industry interviews.

1.1 Thematic literature review

This literature review is not exhaustive, and the article itself cannot mention everything relevant to the paradigm shift in design at large. It looks at design history and research from the service design point of view and focuses on literature that somehow mentions the term shift or turn. Papanek (1971) was perhaps the first to show the variety of social complexities in design. The issues of sustainability and social context must be considered when designing. Buchanan (2001) addressed knowledge production and the epistemologies of design that investigate how we know what we know. He sought to identify the value of design research.

Krippendorff's (2005) book *The Semantic Turn* has been a cornerstone in outlining a new way to look at design as a science. This helped designers review the semantic concerns in design history, present the philosophical background and shift the focus from product functionality to their meaning.

Kimbell (2009) argued for the emergence of service design. Her argument describes how service design can become formally established by becoming part of management and design education curricula and larger corporations' processes and methods. She presents the background to the turn of service design lying in the division of economics between raw materials, manufacturing and services, as well as service design responding to the value co-creation of experience economies that Pine and Gilmore (1999) proposed. Furthermore, she placed the turn to service design within management theories, looking at new ways to produce innovative products and services (Kimbell, 2009). Kimbell's proposal for the new turn did take place. Service design was established both in education and in business. Now, service design is taught in several academic institutions (Becermen & Simeone,

2019; Sarantou & Miettinen, 2022; Suoheimo, Vasques, & Rytilahti, 2020) and multitudes of companies (Accenture, Frog, Futurice, Deutsche Telekom, Volkswagen, Amazon, Netflix, Apple, Starbucks, Zappos, McDonald's, etc.) use it.

Muratovski (2015, 2021) provided background on the shift in the role of design towards a more strategic one. He illustrates a shift in the corporate world that embraces the role of design. He proposes that design's adaptability and the role of education support the evolving field and enable design to play a role in changing economies. This shift expands the role of design across disciplines to support the innovation process taking place almost everywhere. Opazo-Basáez et al. (2022) support Muratowski's idea of a shift from the service innovation point of view. They propose service innovation as a novel technological innovation that enhances performance, speeds up innovation and proposes collaboration across companies.

We can also discuss the ontological turn that Ansari (2019) proposes. He discusses the decolonial turn in design, which aims to bring marginalised designers' voices and concerns into a discussion from an Anglo-Eurocentric perspective. In his opinion, the start of this discussion should be the decolonisation of knowledge systems. The decolonising discussion has many voices in design, especially when addressing participatory research and its ethics (Seppälä et al., 2021).

The sphere of design has become more complex. It also raises questions about the polarities between these different turns. On one hand, design is an approach to speed up and expand technological innovation. On the other hand, the expansion of design demands that we address the questions of ethics, care and dismantling of colonial structures more carefully. Design as systems thinking (Buchanan, 2019) that addresses the design itself as a system consisting of its parts and the world consisting of systems has become more complex to understand (Higgins, 2014).

1.2 Positionality as a second turn in service design

Designers are mediators within a sociotechnical system. Because of this central role, they are also responsible for investigating their positionality and disclosing any power asymmetries in complex social systems (Rodriguez, Schon, & Celi, 2023). They must pay attention to their own biases and privileges and their ways of knowing and doing when undertaking this mediation.

When referring to research, Darwin Holmes (2020, p. 1) describes positionality as a reflection on the researcher's individual worldview or "where the researcher is coming from" concerns, reflecting on individual beliefs about the nature of social reality, the nature of knowledge and the way the researcher interacts with the environment they relate to. The author proposes that positionality unfold in three areas: the research participants, the research context and the subject under investigation.

When applied to service design research and education, the three positionality dimensions suggest some questions: What is the meaning and positioning of design action? Can we imagine service design research and education shifting its focus from individual experience to social mechanisms of change?

1.2.1 Positioning about the participants

An *outsider* positioning of service design education concerning the actors with which they collaborate often provides few elements for describing the participants in a service context (e.g. their needs, preferences, or functional requirements). An *insider* positioning (e.g. working in a context) would provide a deeper understanding of the people with whom a design student is collaborating, their culture and beliefs, and their power dynamics, diversities and social practices. Sangiorgi (2015) proposes a paradigm shift in which design agencies (practitioners) inform design policies and interdisciplinary work, producing specific knowledge that can support social transformation. She argues that all stakeholders can create public value through collaborative design. Furthermore, she argues for systemic change that can take place when operating at a complementary level.

1.2.2 Positioning about the context

While the root of service design in marketing and management disciplines suggests a focus on organisational processes, efficiency and problem solving, the complexity of the context in which service design education operates imposes a more open approach that looks at participatory processes. Consequently, the existing problem- and project-based approaches should be revised in favour of an open-ended *infrastructuring* approach (Björgvinsson et al., 2010) to facilitate social interaction between different practices in a dynamically changing relational context and changing dynamically.

The new landscapes of design initiated the notion of the user as a co-designer (Sanders & Stappers, 2008). Manzini (2015) has developed this notion by considering citizens and everyone's ability to design. The participatory process is more complex. Noronha (2018) proposed a collaborative turn in design. She argues that there is a great challenge in recognising the multiplicity of overlapping worlds and other symbolic systems that promote the equilibrium of communities when doing the action of co-design. Miettinen et al. (2023) further developed this in art and design processes with communities.

1.2.3 Positioning about the subject

Service design education inherits its approach and methods from design disciplines focusing on material objects, be they products, architectures or urban spaces. The outcome of the design process is relatively complete and controllable through a codification process (a blueprint and technical representations). The nature of design intervention in a social context instead has the characteristics of an *enquiry* (Kimbell, 2011), and the results are open transformations at different levels (local interventions as well as policy actions) that cannot be easily controlled or represented by the designer.

1.3 Ethics and values

The expansion of the role of design demands more care. Van den Hoven (2017) discussed the design turn in applied ethics. In the mid-twentieth century, ethics needed to take steps closer to everyday life to be understood and utilised. The challenges coming from the evolution of technology and the designers' cross-disciplinary approach require re-visiting the concepts of agency and responsibility when discussing design, design histories or design agents. Designers themselves outshape the design practice. In this setting, many are already set by actors and previous actors. The global problems (climate change, healthcare, etc.) that we address as designers need more than common-sense morality, as they go beyond particular cultural constructs that may be familiar to us and are beyond individual capacity. Van den Hoven (2017) presents the concepts of "design for values" and "value-sensitive design." These concepts were originally addressed in the human-computer interaction (HCI) discourse by Friedman (1997) at the University of Washington in their Values-Centred Design initiative. Again, we face the question of how values become implicated in technological design, especially when facing the questions of artificial intelligence (AI) and a machine's ability to be an agent.

When addressing design as a system where the participants, the context and the subject play a role, one needs to understand that the new shift is addressing and problematising these positionalities. Understanding the design histories to make sense (Rylander et al., 2022) in this system is relevant.

1.4 Paradigm shift in service design education

Service design education comes from the convergence of design and marketing disciplines. It is the result of the progressive maturation of a branch of design, from the craft and studio-based industrial design in the 1980s to interaction design and human-centred design (HCD; Giacomin, 2014), under the influence of marketing and management disciplines (Normann & Ramírez, 1993; Normann, 2001), up to the development of a systemic perspective, matured through design research that includes social (Manzini, 2015), environmental (Ceschin & Gaziulusoy, 2019), organisational (Cooper et al., 2009) and policy-related (Bason, 2014; Buchanan et al., 2017) issues.

This evolution has paved the way for multidisciplinary service design. Budd (2011) comments on the changes in design education from studio-based learning to exponentially-grown tools that designers are now using. It has become impossible to master all the tools the technologies enable. From his perspective, the change is fast, and design education lags. The article, written in 2011, also references the D-school teaching model combining creative, analytical, and collaborative elements. There has been a shift towards ongoing learning technologies.

a In their article, Meyer and Norman (2020) present major shifts in design education. They focus broadly on HCD. They reference Friedman's (2019) lecture and presentation that proposes that the new context in design is caused by the

challenges that design faces. Meyer and Norman (2020) divide the challenges into four groups: performance challenge focuses on the context of design action, what designers must do and make rather than their skill set.

b Systemic challenges are issues where design addresses large-scale economic, industrial and social frameworks and systems not only parts of them.

c Contextual challenge forces design to address environment, culture and politics and it deals with complex systems.

d Global challenges make us look at the major societal challenges such as Sustainable Development Goals (SDGs).

All these challenges ask for different skill sets, and most probably, designers strengthening their generic skills such as communication, teamwork and learning techniques to address this change. The many critical changes characterising the present historical moment require a deep reflection on the role of design, which should start from the technical, social and cultural grounds underpinning design education. Eggink (2021) proposes a strong interplay between design research and the philosophy of technology in design engineering studies. He presented an idea of the practical turn for applying ethics while teaching the responsible design workshop. The workshop creates a living laboratory for analysing and applying principles of responsible design proposed strongly in the philosophy of technology. He suggested that the practical context helps in understanding and applying philosophy.

The need for a paradigm shift in design education has been discussed by several authors over the years from different disciplinary perspectives (De Parker, 2013; Ranjan, 2010), often focusing on the skills and competencies that new designers should acquire throughout their education (Noweski et al., 2012; Rocha et al., 2018) and the need to change our educational practices accordingly (Adams et al., 2003; Carvalho & Goodyear, 2018). On the other hand, service design education has predominantly emphasised designers as problem solvers. At the same time, the proposition of a new form of societal involvement necessitates that upcoming designers grasp and navigate conflicts, tensions, agonism and complexity (Ehn, Nilsson et al., 2014).

The existing literature discusses the need for a paradigm shift, clearly highlighting the inadequacy of the existing paradigm in addressing emerging problems. However, the authors of this chapter proposed exploring how the need for change has been perceived and metabolised in the practice of service design and service design education. This led to a double investigation—through a workshop with service design educators and many interviews with service design practitioners.

2 Research methods

The research process is based on reflexive dialogical research, in which collaborative dialogue is supplemented with reflexive practice when writing theory (Hibbert et al., 2014). Hibbert et al. (2014) suggest a theory-building stage with pre-research

conceptualisation that includes the latent resources for theory building. As all of the authors of this chapter are service design scholars, it is obvious that there are a number of latent resources. After this, there are emerging theorisations while conducting research. Here are the different phases of the research process: (a) literature review, (b) co-design workshop and (c) expert interviews, all impacting emerging theorisations. Finally, there is a refinement of theory with a larger context. This approach includes both self-reflection and critical reflection through the process (Boyd et al., 2022). The authors had collective discussions throughout the process, combined with notes and article writing (Ripamonti et al., 2016).

The desktop literature review was conducted by searching for the most relevant publications in the field without limiting them to a certain journal or geographical region (Webster & Watson, 2002). One downside of literature reviews is that they are bound by time. The results from one day to another can differ. Affinity diagrams (Lucero, 2015) were used for the workshops and interviews. All three types of data collected aimed to find themes (Clarke & Braun, 2017); a cross-check found themes via method triangulation (Carter et al., 2014). In qualitative research, it is good to consider how the researchers' backgrounds and worldviews can influence the results (Creswell, 2009).

2.1 Co-design workshop

The workshop participants included researchers and educators in service design or related fields, such as experience or interaction design. The 16 participants were heterogeneous, presenting various educational levels, nationalities and universities.

During the workshop, the participants were divided into four groups to discuss three different themes in 15-minute sessions related to the paradigm shift in service design education. One of this chapter's authors facilitated the workshop, and two facilitated the discussion in two groups. The theme proposed to them was *if and how service design education can undertake a paradigm shift to address the complexity and challenges of contemporary society*. A radical change was initially proposed to the participants as a challenge that could lead to a discussion about the deep transformation of the existing system.

The three dimensions for the proposed repositioning were introduced to form the structure of the three workshop sessions. The first theme discussed the positionality of the participants. The second topic focused on service design education. The third theme concentrated on a larger focus on the mechanisms of social interaction or policymaking, which implies a deeper reflection on how to support change and radical transitions in social systems (communities or society). Each group wrote their ideas on Post-it notes during the workshop and placed them on a sheet. There was a five-minute discussion with the whole group after each session, in which each group presented their findings. The Post-it notes were analysed using the affinity diagram method (Lucero, 2015) and divided into themes (Clarke & Braun, 2017) on a Miro board, as shown in Figure 4.1.

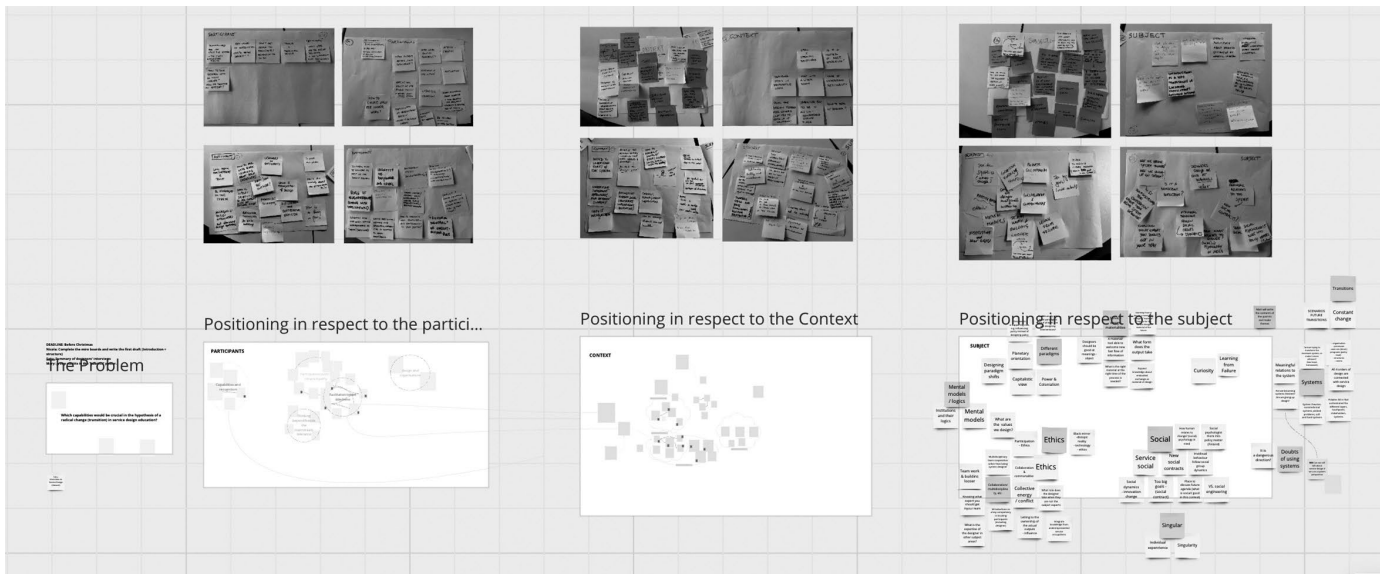


FIGURE 4.1 Clustering themes on a Miro board.

2.2 Interviews with senior service designers and managers during the Empathy Business project

For the Empathy Business research project (2023–2024), 20 interviews were conducted with senior service designers and managers during the spring of 2023. These interviews focused on developing digital tools for service designers. The strong focus in the interviews was on developing the future of service design as a practice and investigating the skill sets needed by the service designers and the topics that the service designers deal with in their daily work (Miettinen et al., 2024). There has been a significant change in working life since COVID-19. The post-COVID working life is more digitalised, and distributed online work has become part of daily life (Vyas, 2022). These requirements and jump in digitalisation outline the tools needed for service design.

The interviews had a pre-set of questions focusing on several clusters of different topics: service designers' workflow with digital tools and description of needs for these, questions of sustainability in designers' work and more strategic role of service design and its impacts. This created a structure for the interviews. The interviewer left sufficient room for the interviewee's intuition and experience and respected the choice of topics that the interviewee wanted to discuss. This set the tone of the interviews.

The interviewees represented a group of designers in Europe, Latin America and Asia. They could be categorised into three groups

a working as a senior in-house designer in a major corporation,
b working as a senior or partner in a medium-sized design consultancy, and
c working as a senior designer in a small design consultancy.

The recorded interviews were analysed using the affinity diagram method (Lucero, 2015) and thematic analysis (Clarke & Braun, 2017); handwritten notes were made directly after the interviews. The interview recordings were transcribed with the help of AI. After this, the interviewers produced affinity diagram notes that were exported to a Miro board. The notes included key findings from the interviews. The research group worked collectively to categorise and cluster the notes and create an affinity diagram. Figure 4.2 shows the rounds of analysis of the interviews.

3 Findings

3.1 Literature review

We learned from the literature that service design education has a strong user-centred tradition. Still, in practice, when working on a participatory service design project, the subject is often a community (Meroni, 2007) and a "larger system of actors," such as ecosystems (Vink et al., 2017).

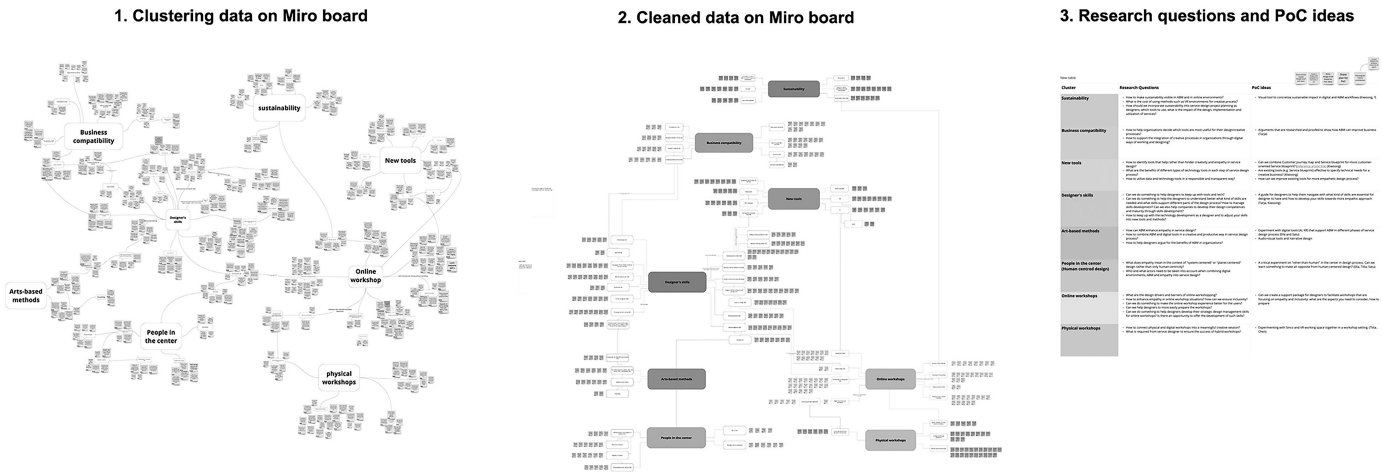


FIGURE 4.2 Affinity diagram on the Miro board used to analyse the interviews (Image credit: Satu Miettinen).

A combination of service design and "transition design" (Suoheimo, 2020) has been considered a relevant aspect of systemic opening design education. This discussion introduces the perspective of a "paradigm shift" that overturns the supporting role that design has had so far to a capitalistic view and forms of economic, social and cultural power. This paradigmatic change originates from a critical perspective on the idea of the designer as part of the problems caused by the climate crisis and globalisation (Morelli, 2012), and it becomes more weighty and urgent with the actions of the movements for the decolonisation of design (Smith et al., 2020).

Ethical issues that deal with the power dynamics and responsibilities of the designers arise. Working in collaboration with others also requires a deep understanding of "ethical issues" that may emerge, especially when considering perspectives such as feminist or decolonising thinking or the inclusion of disadvantaged groups. In this landscape, service design education should reposition itself in the new "decolonial and pluriversal context" (e.g. by recognising cultural plurality) (Duan et al., 2021; Seppälä et al., 2021). This is particularly urgent in light of the diffusion of service design education to countries and cultures, where the present sociotechnical paradigm, based on the dominance of Western culture, is revealing all its weaknesses (Duan, 2024). Scholars in the Global South have proposed a paradigm change to address inequality and to propose plurality for some time (Mignolo, 2009; Tejeda, Espinoza, & Gutierrez, 2003).

3.1.1 Workshop findings

The input for the first discussion session focused on participants (i.e. the people interacting with service design education) and the shift of the designer's role and the role of education from an *outsider* to an *insider* and co-participant position. The participants' discussion highlighted different issues summarised in the following subparagraphs.

3.1.2 Positioning with respect to the participant

The new role designers can have as *insiders* when working with communities or in public services implies that service design education also stresses the need to respect the people in their context, avoiding any extractive approach but learning how to facilitate interaction dynamics. This means that service designers are recommended to:

Focus on communities rather than individual "users" or "citizens": The workshop confirmed the need for a radical shift from a user-centred perspective to a community-related one. The participants discussed many of the issues related to this shift, including the need for a different way to "enter" the community, to consider them as partners instead of a source of information, and consequently to consider not only what the students can get from the community but also what they

can give back. Issues related to power relationships have been stressed as very relevant, especially in disadvantaged social contexts.

Consider the different competencies of the participants: When working among people, different competencies play a role. Some people will contribute highly specialised knowledge, and others will have expertise in managing their practices, traditions, and ways of doing and interacting with each other. The competencies a design education can offer (and simultaneously acquire by working in this context) concern both hard technical skills and soft skills, such as mediation, facilitation, trust building and critical thinking.

Ensure neutral facilitation: Issues emerge concerning the position of the facilitator, which may be external to the system (as an outsider) or embedded in it, and, with respect to those positions, it would also be essential to discuss the need for the facilitator to be neutral with respect to the issues to work on. The facilitator should also be able to consider different perspectives and highlight differences in participants' thinking and practices, not necessarily seeking opposition but considering the differences. Working in collaboration with others also requires a deep understanding of "ethical issues" that may emerge, especially when considering perspectives such as feminist or decolonising thinking or the inclusion of disadvantaged groups.

3.1.3 Positioning with respect to the context

The second session of the discussion dealt with the context in which service design education operates now and should/could operate in the future, moving the focus from problem solving and local solutions to systemic change. Service design education should consider the complexity of this scale of action. Therefore, the following themes were identified:

The context as a system: Different aspects of this condition have been discussed. They include an approach to wicked problems and the student's capability to consider or recognise the need for actions at different levels, from individual behaviours to small social groups (micro-scale) to infrastructure and policy systems (meso scale) and large cultural landscapes (macro-scale).

Evaluating the effects of design action: It is crucial and requires understanding different system components and figuring out elements for a quantitative or qualitative evaluation.

Teaching complexity and building the necessary capabilities: education should support students in making sense of complexity without overwhelming them. Students should be able to deal with uncertainty, for instance, by addressing the different system levels (structures, behaviours, etc.) separately.

Understanding power dynamics: It means recognising and possibly addressing the inevitable asymmetries and vulnerabilities that emerge in a participatory context. One group of participants discussed the need for design education to *be invited* into the context rather than proposing interaction with local citizens as a mere source of information.

3.1.4 Positioning with respect to the subject

The last session of the workshop dealt with the subject, or the *material,* of design action (Blomkvist, 2014), which, in service design, is shifting from the interaction between services and their users to the mechanisms of social interaction, the structure of public services and support for policymaking. Therefore, the discussion focused on supporting change and radical transitions in social systems (communities or society).

The shift towards a "system approach" has been discussed from different perspectives. On one hand, it has been observed that service design's domain (the *material* it manipulates) has expanded and is now covering all orders of design (Buchanan, 1992) and will, therefore, require service designers to learn how to orchestrate systemic components. On the other hand, the limits to service designers becoming system thinkers have been discussed—whether this expansion is going towards a different professional domain and, therefore a substantially different education.

One significant topic covered was the "social value" implied in the direct interaction of design education in real contexts. In particular, a need emerged to recognise the systemic results of social interactions as different and noncomparable with the sum of individual experiences of a service.

The contribution of service design to "transitions" has been considered a relevant aspect of the systemic opening that design education should face. This discussion introduces the perspective of a "paradigm shift" that overturns the supporting role that design has had so far to a capitalistic view and forms of economic, social and cultural power. This paradigmatic change originates from a critical perspective on the idea of the designer as part of the problems caused by the climate crisis and globalisation, instead of being part of the problem solution (Morelli, 2012).

3.2 Interview findings

The roles and job descriptions of service designers have become more varied. Designers need a very "broad spectrum of generic skills." This can also mean having a strong foundation in mathematics, engineering and history. Mathematical skills are very handy when making sense of probabilities and the scale of things and understanding the business side of services. Engineering skills facilitate the building of different prototypes and demos at the very core of digital service design.

Having a broad spectrum of skills makes it possible to operate as an agent in the design system and to analyse this front from a "holistic view." Service designers are crossing organisational boundaries and working in multidisciplinary teams that address "political topics" and need even to understand different legislation. The workload is distributed in international organisations, and information needs to be sourced from different channels. It seems evident that designers need to understand the world holistically and have strong analytic abilities to make sense of it. They

must be able to use tools such as mapping to analyse "relationships and interactions between different systems." This proposes that designers utilise design to analyse the world and understand what kinds of systems exist and how they interact.

The paradigm for "sustainable change" proposes that designers need sustainability tools and understanding. Tools are needed to help the designer evaluate what is sustainable. There is a call for sustainability and responsibility in strategic service design. For example, one must consider the need for travel and the use of resources. Sustainable development should be at the core of the service designers' toolkit. There must be a wide understanding of sustainability and what must be considered according to all 17 SDGs of the United Nations Department of Economic and Social Affairs (UN DESA, 2023). Designers' specific competence is to understand the product life cycle and the idea of circularity. An important aspect of sustainability is the social aspect and wellbeing of employees. Designers must understand the discourses and terms prevalent in "business" and have "strategic leadership and planning skills." The more designers work in management positions, the more they need to understand companies' processes, goals and strategies. Design managers can see a holistic view of the business. The strategic management process is not the same as the service design process. Designers must have strategic thinking, management and business modelling skills. There is a need for learning strategic business language and embedding it into design studies. When designers have strategic thinking skills, it is easier to imagine possible futures, understand goals (of the company, customers and service) and map ways to get there. Of course, strong service design skills increase the ability to evaluate service access, review services and understand how they bring value to customers and the company.

4 Discussion

Similar themes found in the data collection are summarised in Table 4.1. The contribution of service design to "transitions" has been considered a relevant aspect of the systemic opening that design education should face. This discussion introduces the perspective of a "paradigm shift" that overturns the supporting role that design has had so far to a capitalistic view and forms of economic, social and cultural power. This paradigmatic change originates from a critical perspective on the idea of the designer as part of the problems caused by the climate crisis and globalisation (Morelli, 2012), and it becomes weightier and more urgent with the actions of the movements for the "decolonisation of design" (Smith et al., 2020).

The workshops and interviews discussed in this paper present two perspectives relevant to service design education. These two perspectives describe the capabilities of service design education inside and outside the existing paradigm. While the interviews focused mainly on designers and practitioners working in the private sector, many workshop participants were instead researchers working in the public

TABLE 4.1 Similar themes found in different data collections

Literature review	*Workshop*	*Interviews*
Transitions	Transitions	Strategic skills
Ethical issues	Ethical issues	Sustainable change
	Understanding power dynamics	Political topics
	The context as a system/Evaluating the effects of design action	
	Ensure neutral facilitation	
	Consider the different competences of the participants	
Larger system of actors	System approach	Relationships and interactions between different systems
	Teach complexity and build the necessary capabilities	Holistic view
Decolonial and pluriversal context	Focus on communities rather than individual "users" or "citizens"	
	Social value	
Paradigm shift	Paradigm shift	
		Business
		Broad spectrum of skills, generic skills

sector or with a specific focus on social innovation. Interviews, workshops and literature reviews highlight the "systemic aspects" as an urgent topic to address in service design education. Still, in the interviews, the practitioners highlighted operational and organisational issues related to the present paradigmatic framework, thus suggesting an expansion and improvement of the paradigm for design education to include "business, organisational and strategic" aspects (and many design educations are already addressing this issue).

An extended view beyond the existing paradigmatic conditions for service design education would highlight issues that the existing methodological approach in service design education may not be able to address. Furthermore, the domain of service design overlaps and merges with "systemic design, policy design and social innovation," and it is impossible to define one specific or "pure" set of competencies related to service design, since the different perspectives are all connected. This was also suggested by the multiple paradigms or logics noted in the workshop's Post-its, confirming that working with complex problems and systems requires a more "holistic approach," which could include "multiple paradigms" and "epistemologies" (Midgley, 2000).

The radical and complex nature of the transition makes it hard to describe a theory of change in the practice of service design and service design education. However, the interviews and the workshop tried to highlight the directions for a change towards a systemic transition.

In both cases, service design must define a solid set of skills and capabilities to work on a systemic view of the context and content of the action. The material or the context for design action moves "from interactions to (sociotechnical) systems" (Ceschin & Gaziulusoy, 2019; Rylander & Eneberg, 2023). On the other hand, defining a direction requires a stronger capability of designers to read and understand how sociotechnical systems capture or facilitate the emergence of the most desirable directions from different social instances, expressed by the diversity of stakeholders, practices, cultures and economic conditions present in a context. The work service designers are requested to do for the participants is related to a view of the system not as a uniform material but as a "heterogeneous and pluriverse entity," in which the definition of the directions of change is often controversial and cannot ignore the cultural, social and economic differences that all the participants are bringing in. Compared to the workshop, the interviews emphasised the business knowledge of service design more.

A clear indication of a new *praxis* for service designers concerns the role and action of service designers in systemic processes of change. In this process, the role of designers can no longer be to generate solutions, but rather to facilitate a co-creation process. Design education should consider alternating project-based activities (i.e. controlled processes with a well-defined brief, a defined process timing and a "finished" solution) with open-ended activities in which the design students learn how to facilitate, provoke or trigger processes of change that escape from the designer's control (Hillgren et al., 2011). This could even imply a different organisation of service design education, as open-ended work would challenge the traditional project-based structure of many programmes or even the semester-based time structure. While this change may exceed the adaptability of several education systems, it may also suggest a new interaction within service design education, which establishes continuous interaction between design education and a "social context" (Fassi & Manzini, 2022; Thorpe & Rhodes, 2018).

Our findings go in part hand in hand with the proposed changes by Meyer and Norman (2020) and Friedman's (2019) lecture for design education that stresses the need for understanding performance that could be understood as business skills, systems, the contextual challenges and global challenges as part of the systems; some systems are global, such as sustainability in supply chain management services.

We recognise this study's limitations and how, for example, we could have selected other themes discussed in the workshop. We used the lens of participant, context and subject and understood that using others could lead to other results. In addition, other data collection methods could yield different outcomes.

5 Conclusion

In its maturation, service design thinking has expanded its complexity, thus including different or sometimes even divergent directions. The results of the workshop and the interviews show that service design thinking is still exploring areas of interest in the existing sociotechnical paradigm. At the same time, it is contemplating

the possibility of working outside this paradigm and exploring areas not fully explored by service design, even though they are the common ground of research for other disciplines, such as social science, anthropology and cultural studies. This could lead to embracing more paradigm incommensurability and using multiple paradigms to have a better and more holistic view of the challenge at hand.

From a systemic view, service design can be investigated as a design system. This helps us understand how to develop design education, and the interviews shed light on this. We recognise that it would require more investigation into what kind of service design system is. Positionality can also help analyse different systemic stances that are complex and plural; this complexity creates pressure for education to keep up with the speed of transformation and change.

This means that there are different avenues for the evolution of service design capabilities, and they may not necessarily belong to a unified profile of service designers. The complexity of the service design discipline has possibly matured to a point that needs a different articulation of competencies, including a critical review of the existing framework for service design education.

Acknowledgements

The study is part of the Business Finland-funded RRF project "Empathy Business: How to Digitalise Service Prototyping and Business through Creativity," with grant agreement No. 7425/31/2022 from 01.11.2022 to 31.12.2024. We wish to thank all the participants in the workshop at the Servdes conference in Rio 2023 and the participants in the interviews.

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